

## Cyclotomic Extensions

Splitting field of  $x^n - 1$  over  $K$  and we are interested in the cases where  $\text{char}(K) = 0$  or  $\text{char}(K) = p$ ,  $p \nmid n$ .

**Theorem:** Let  $n \in \mathbb{N}$ ,  $K$  be a field such that  $\text{char}(K) \nmid n$ , let  $F$  be a cyclotomic extension of  $K$  of order  $n$ .

(i)  $F = K(\zeta)$  where  $\zeta \in F$  is a primitive  $n$ th root of unity.

(ii)  $F$  is an abelian extension of dimension  $d \mid \varphi(n)$

(iii)  $\text{Aut}_K(F)$  is isomorphic to a subgroup of  $\mathbb{Z}_n^\times$ .

**Proof:** Let  $F$  be the splitting field of  $x^n - 1$  over  $K$ , and let  $\zeta$  be a primitive  $n$ th root of unity. The roots of  $x^n - 1$  are  $\zeta, \zeta^2, \dots, \zeta^n$ , so  $F = K(\zeta, \dots, \zeta^n) = K(\zeta)$ .

(iii)  $\Rightarrow$  (ii), subgroups of abelian group are abelian.

To prove (iii), if  $\sigma \in \text{Gal}(F/K)$ ,  $\sigma(\zeta)$  of some root of  $x^n - 1 = \text{some } \zeta^j$  for  $j \in \mathbb{Z}_n$ . Since  $\sigma(\zeta)$  determines  $\sigma$ , this

gives a function  $\theta : \text{Gal}(K(\zeta)/K) \rightarrow \mathbb{Z}_n$ . Note, if

$\theta(\sigma) = j$  and  $\theta(\tau) = m$ , then  $\sigma\tau(\zeta) = \sigma(\zeta^m) = \zeta^{jm}$ , so

$\theta(\sigma\tau) = \theta(\sigma)\theta(\tau)$  in  $\mathbb{Z}_n$ . So  $\theta$  is a group homomorph.

$\ker(\theta) = \{ \sigma : \sigma(\zeta) = \zeta^{-1} \} = \{ \text{id} \}$ , so  $\text{Gal}(K(\zeta)/K) \hookrightarrow \mathbb{Z}_n^\times$

**Note:** We might have  $\zeta \in K$  so  $\text{Gal}(K(\zeta)/K)$  might be trivial.

**Definition:**  $\Phi_n(x) = \prod_{d|n} \Phi_d(x)$

•  $n = 1$ ,  $\Phi_1(x) = \prod_{d|1} \Phi_d(x) = x - 1$

•  $n = 2$ ,  $\Phi_2(x) = \prod_{d|2} \Phi_d(x) = x + 1$

- $n=3$ ,  $\Phi_3(x) = x^2 + x + 1$

- $n=p$  (prime),  $\Phi_p(x) = x^{p-1} + x^{p-2} + \dots + 1 = \frac{x^p - 1}{x - 1}$

**Proposition:**  $\Phi_n(x)$  is irreducible over  $\mathbb{Q}$  for all  $n \in \mathbb{N}$

and the splitting field of  $x^n - 1$  has Galois group  $x^n - 1$

**Proof:** For  $p$  prime, note  $\Phi_p(x+1) = x^{p-1} + px^{p-2} + \dots + p$

so by Eisenstein,  $\Phi_p(x+1)$  is irreducible over  $\mathbb{Q}$  of degree  $p-1$ .

so  $\text{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q}) \simeq \mathbb{Z}_p^\times$ .

**Theorem:** (Kronecker - Weber) Every abelian extension of  $\mathbb{Q}$  is contained in a cyclotomic extension of  $\mathbb{Q}$ .

## Radical Extensions

**Definition:** An extension field  $F$  of  $K$  is a radical extension of  $K$  if  $F = K(\alpha_1, \dots, \alpha_n)$  where some power of  $\alpha_1$  is in  $K$  and for each  $k \geq 2$ , some power  $n_k$  of  $\alpha_k$  lies in  $K(\alpha_1, \dots, \alpha_{k-1})$ .

**Question:** When is  $K(\alpha)$  or the splitting field of  $f(x)$  a radical extension? Contained in one?

**Definition:** Let  $K$  be a field and  $f(x) \in K[x]$ . The equation  $f(x) = 0$  is solvable by radicals if there exists a radical extension  $F$  of  $K$  and a splitting field  $E$  of  $f$  over  $K$  such that  $K \subset E \subset F$ .

**Lemma:** If  $F/K$  is separable extension by radicals, then the Galois closure of  $F$  is also an extension by radicals.

**Proof:** A radical extension of a radical extension is a radical extension. Let  $F = K(\alpha_1, \dots, \alpha_r)$  as in the definition. By def,  $\alpha_1^{m_1} \in K$  for some smallest  $m_1 > 0$ . Let  $K_1$  be the splitting field of  $x^{m_1} - \alpha_1^{m_1}$  over  $K$ . This is a radical extension because every root  $\beta$  of  $x^{m_1} - \alpha_1^{m_1}$  satisfies  $\beta = \alpha_1^{m_1}$ , so  $K_1$  is a radical extension of  $K$  and  $K_1$  is contained in the Galois closure of  $F$ , because  $\alpha_1 \in F$ . Inductively proceed...  $K_2$  is the splitting field of the min. polynomial of  $\alpha_2$  over  $K_1$ . Eventually

$F \subseteq K_r \subseteq \overline{\text{Gal}}$ , so  $K_r$  is Galois,  $K_r = \overline{\text{Gal}(F)}$  but  $K_r$  was constructed as an extension by radicals.

**Theorem:** Let  $F/K$  be an extension by radicals, and let  $K \subseteq E \subseteq F$  with  $E/K$  Galois. Then  $\text{Gal}(E/K)$  is a solvable group. So in particular, if  $f(x) \in K[x]$  is solvable in radicals, then the Galois group of  $f$  is solvable.